

DualNet: A dual-path network with adaptive compensation for multivariate time series forecasting

Huashun Li ^{*}, Weimin Wu, Wei Chen, Mei Zhang

School of Electrical and Information Engineering, Anhui University of Science and Technology, 168 Taifeng Street, Huainan, 232001, Anhui Province, China

ARTICLE INFO

Communicated by F.G. Guimaraes

Keywords:

Time-series forecasting
Residual learning
Temporal dynamics
Multivariate prediction
Spatial decomposition

ABSTRACT

Multivariate time series prediction remains a fundamental challenge across various domains due to its complex temporal dynamics and inherent dependencies among variables. This paper introduces DualNet, an innovative architectural paradigm based on decoupling principles, which achieves effective decomposition and complementary enhancement of prediction tasks through spatial decomposition. The architecture synergistically integrates global pattern extraction and adaptive local compensation mechanisms, establishing two independent yet complementary learning spaces through functional decoupling. Our framework introduces a dual-path learning strategy that decomposes the prediction process into complementary components: a primary trajectory estimator focused on capturing inherent temporal evolution patterns, and a dedicated compensation mechanism that performs fine-grained calibration through residual learning. This decoupled design not only reduces the learning complexity of individual modules but also enhances overall prediction performance through complementary effects. We developed a temperature-scaled adaptive weighting scheme that dynamically adjusts compensation intensity based on temporal context, enabling the model to achieve optimal balance between prediction stability and adaptive refinement. Through the dual mechanisms of spatial decoupling and complementary enhancement, the architecture achieves organic unification of global feature extraction and local fine-tuning, while incorporating novel normalization strategies and hierarchical feature transformation mechanisms to enhance the model's representational capacity. Comprehensive experiments conducted on various benchmark datasets demonstrate that this decoupling-based dual-channel architecture significantly improves the model's capability to capture complex temporal patterns while maintaining prediction stability. The code is available at <https://github.com/ZS520L/DualNet>.

1. Introduction

Multivariate time series forecasting and spatio-temporal data analysis play crucial roles in modern data science, with widespread applications in traffic flow prediction, human motion prediction, urban computing, and intelligent transportation systems [1–5]. However, accurate long-term prediction remains a significant challenge due to the inherent complex temporal dynamics and inter-variable dependencies in time series data [6,7]. While deep learning models have made remarkable progress in this field, there is still room for improvement in terms of prediction stability and accuracy for long-term forecasting [8,9].

The design of this work is motivated by two key observations regarding time series forecasting. First, different feature channels often exhibit varying temporal patterns and dependencies, making unified processing suboptimal for capturing the diverse characteristics inherent in multivariate time series data [10,11]. Second, the importance of

global and local patterns varies across different prediction horizons and feature types, suggesting the need for a more adaptive and flexible modeling approach that can effectively balance both temporal scales [12,13]. Moreover, in time series forecasting, sequences often display unique local characteristics at points of dramatic changes (such as mutations, seasonal transitions, or anomalous events) [14]. Traditional approaches, which heavily rely on global patterns, struggle to capture these critical local variations accurately. This underscores the necessity of enhancing local compensation mechanisms for improved model performance at crucial timestamps.

Recent years have witnessed significant advances in deep learning for time series forecasting, and various neural network architectures have emerged. CNN-based models (e.g., MCNN [15], TCN [16]) excel at capturing local temporal patterns. RNN-based models (e.g., LSTM [17], GRU [18]) specialize in modeling sequential dependencies, while Transformer architectures (e.g., Informer [19], Autoformer [20]) achieve efficient modeling of long-term dependencies through self-attention

^{*} Corresponding author.

E-mail address: 2021200797@aust.edu.cn (H. Li).

<https://doi.org/10.1016/j.neucom.2025.130720>

Received 13 December 2024; Received in revised form 18 May 2025; Accepted 4 June 2025

Available online 24 June 2025

0925-2312/© 2025 The Authors. Published by Elsevier B.V. This is an open access article under the CC BY license (<http://creativecommons.org/licenses/by/4.0/>).

mechanisms. Recently, studies have shown that even simple MLP structures (e.g., TSMixer [21], DLinear [8]) can achieve competitive performance in certain time series forecasting tasks. However, these models tend to approach the prediction task as a unified end-to-end architecture, lacking explicit consideration of the inherent multi-scale characteristics in time series [22]. Particularly in multivariate prediction scenarios – where models need to simultaneously handle temporal evolution patterns and complex inter-variable dependencies – such unified processing approaches may not fully utilize the complementarity of information at different scales [23–25].

Through in-depth analysis of existing methods, we find that the key to time series forecasting lies in balancing the capture of global trends with local detail refinement. Existing models often employ a unified architecture to handle both aspects simultaneously, which may result in one aspect's performance being constrained by the other [26,27]. For instance, increasing sensitivity to local fluctuations might compromise the accurate capture of global trends, and vice versa. Moreover, existing methods typically use fixed weights to integrate features at different scales, a static fusion strategy that struggles to adapt to the dynamic nature of time series [28,29].

To address these challenges, we propose DualNet, an innovative dual-path learning framework that enhances prediction performance by decoupling global pattern extraction from local compensation mechanisms. Specifically, DualNet comprises two complementary core components: a primary path responsible for capturing inherent evolution patterns in time series, and a specialized compensation mechanism that achieves fine-grained calibration through learned error distributions. To optimize the synergy between these components, we design an adaptive temperature-scaled weighting scheme that automatically adjusts compensation intensity based on temporal context, thereby achieving balance between prediction stability and adaptive refinement. Notably, the architecture is particularly effective for multivariate series where features naturally form related groups (e.g., different sensor types or geographical locations) and when local variations are critical for downstream applications.

The main contributions of this paper can be summarized as follows:

- We propose DualNet, a dual-path framework that achieves enhanced prediction performance by decoupling global pattern extraction and local compensation mechanisms.
- We design an efficient global feature extraction strategy that strengthens the modeling of long-term dependencies in time series via dynamic feature space mapping.
- We develop a temperature-scaled dynamic weighting scheme that adaptively adjusts compensation intensity based on temporal context, enabling dynamic prediction optimization.

The remainder of this paper is organized as follows. Section 2 reviews related work in time series forecasting and deep learning approaches. Section 3 presents the detailed methodology of DualNet, including its architectural design and theoretical foundations. Section 4 provides comprehensive experimental results and analysis. Finally, Section 5 concludes the paper and discusses future research directions.

2. Related work

In this section, we review related work relevant to our research, primarily focusing on recent advances in multivariate long-term time series forecasting. We organize the related work into several subsections based on different methodologies and model architectures.

2.1. CNN-based methods

Convolutional Neural Networks (CNNs) have demonstrated their unique advantages in time series analysis through their ability to capture local patterns and multi-scale features efficiently. MCNN [15] pioneered the multi-scale approach in time series classification by introducing a novel multi-branch architecture that automatically extracts features at different scales and frequencies. MICN [30] combines local features and global correlations through a multi-scale branch structure, utilizing down-sampled and isometric convolutions to achieve superior performance while maintaining linear complexity. SCINet [31] leverages the temporal preservation property of downsampled sequences, introducing a recursive downsample-convolve-interact architecture that effectively captures complex temporal dynamics. PSTA-TCN [16] further advances this direction by combining parallel spatio-temporal attention mechanisms with stacked TCN backbones, enabling stable predictions over extended time horizons while maintaining computational efficiency.

The evolution of CNN architectures has further led to innovative designs that combine the strengths of different deep learning paradigms. TSLANet [32] introduces an innovative Adaptive Spectral Block that incorporates Fourier analysis to enhance feature representation and handle both long-term and short-term dependencies while effectively managing noise. TimesNet [9] proposes a novel approach by transforming 1D time series into 2D tensors based on multiple periods, enabling better modeling of intraperiod and interperiod variations through 2D kernels. ConvTimeNet [33] presents a hierarchical fully convolutional approach with adaptive segmentation and integrated deepwise and pointwise convolutions, successfully combining the advantages of both CNN and Transformer architectures. These developments showcase how CNN-based models continue to evolve, offering efficient and effective solutions for time series analysis through their ability to capture multi-scale temporal patterns and maintain computational efficiency.

2.2. RNN-based methods

Recurrent Neural Networks, particularly LSTM variants, have established themselves as powerful tools for capturing temporal dependencies in time series forecasting. LSTNet [17] pioneered the integration of CNN and RNN architectures to simultaneously capture both short-term local dependencies and long-term patterns, while incorporating traditional autoregressive components to address scale sensitivity issues. DeepAR [34] introduced a probabilistic forecasting approach using autoregressive recurrent networks, demonstrating the capability to generate accurate probability distributions for future predictions by training on multiple related time series. GLSTM [35] further advanced the field by combining LSTM with genetic algorithms, where GA optimizes crucial hyperparameters such as window size and neuron count, resulting in significant improvements in prediction accuracy.

The evolution of LSTM-based architectures has led to increasingly sophisticated and practical applications in various domains. Short-term load forecasting research has demonstrated LSTM's superior capability in handling complex electrical load predictions with multiple exogenous variables, outperforming traditional statistical methods across different time horizons [36]. The development of optimized LSTM models has further enhanced forecasting capabilities by incorporating automated hyperparameter selection through grid search methods, enabling these models to effectively capture nonlinear patterns in non-stationary time series data while maintaining robust performance across diverse forecasting scenarios [37].

2.3. Transformer-based methods

In recent years, Transformer-based time series forecasting models have made significant progress, with researchers making various improvements and innovations to the original Transformer architecture to better adapt to the specific requirements of time series forecasting tasks.

Early improvements mainly focused on enhancing the model's ability to handle long sequences. Informer [19] introduced the ProbSparse self-attention mechanism, reducing computational complexity from $O(L^2)$ to $O(L \log L)$, and designed a self-attention distillation mechanism to handle extremely long input sequences. Autoformer [20] breakthrough by incorporating series decomposition as a basic internal module of deep models and designed an auto-correlation mechanism based on time series periodicity, achieving significant improvements in long-term forecasting tasks across multiple domains. ETSformer [38] drew inspiration from classical exponential smoothing methods, proposing exponential smoothing attention and frequency attention to replace traditional self-attention mechanisms, while introducing interpretable modular decomposition structures.

Recent research has placed greater emphasis on modeling relationships between multivariate time series. Crossformer [39] effectively captures both temporal and variable dimensional dependencies through innovative dimension-segment-wise embedding and two-stage attention layers. iTransformer [40] reconsidered the traditional Transformer architecture by applying attention mechanisms and feed-forward networks on transposed dimensions to better learn variate-centric representations. The Client [41] model employs linear modules to learn trend information while utilizing attention modules to capture cross-variable dependencies, achieving excellent prediction performance while maintaining low computational costs. FEDformer [42] combines seasonal-trend decomposition methods with frequency domain enhancement, improving long-term prediction capabilities while maintaining linear complexity.

2.4. MLP-based methods

Recent studies have shown that while Transformers have achieved notable results in time series forecasting, their high computational complexity and potential loss of temporal information have led researchers to explore lightweight MLP-based architectures. DLinear [8] demonstrated through experiments that simple linear models can outperform complex Transformer models in long-term time series forecasting tasks, providing new research directions for the field. FITS [43] innovatively operates through interpolation in the complex frequency domain, achieving comparable performance to existing models with only about 10,000 parameters. Inspired by MLP-Mixer in computer vision, TSMixer [21] incorporates innovative designs such as online reconciliation heads and hybrid channel modeling, surpassing Transformer models in prediction accuracy. TimeMixer [44] introduces a new paradigm of multiscale mixing, utilizing PDM and FMM modules to process temporal patterns at different scales, achieving significant results in both long-term and short-term prediction tasks. SOFTS [45] proposes the STAR module, adopting a centralized strategy to handle inter-channel interactions, achieving excellent prediction performance while maintaining linear complexity. In addition, COLLAR [46] proposes an innovative architecture with dynamic Koopman regularization for high-dimensional multivariate time series forecasting, which effectively models the linear evolution of multivariate time series by capturing nonlinear temporal structures within a low-dimensional latent subspace. These works collectively demonstrate that carefully designed MLP architectures can not only reduce computational complexity but also achieve performance superior to Transformers in time series forecasting tasks.

Inspired by these studies, we recognize that in the field of time series forecasting, simple and efficient architectural designs often achieve

outstanding performance. In particular, the success of DLinear and FITS demonstrates that through rational model design, prediction effectiveness can be maintained or enhanced while reducing parameter count. Based on this insight, we propose a novel dual-path neural network architecture, DualNet. This architecture, through spatial decoupling, decomposes complex prediction tasks into two complementary learning paths: one focused on capturing the main evolutionary trends of time series, and the other dedicated to fine-grained local feature modeling and error compensation. This design philosophy echoes the approaches of TSMixer and TimeMixer in multi-scale feature extraction, but we go further by introducing an adaptive weighting mechanism that allows the model to dynamically adjust the contribution ratio of both paths according to prediction needs at different time scales. Meanwhile, drawing from SOFTS's experience in channel interaction, we designed efficient feature fusion modules that achieve more effective multivariate time series modeling while maintaining linear computational complexity growth. Experimental results show that this decoupling-based dual-path architecture not only inherits the computational efficiency advantages of lightweight models but also achieves significant improvements in prediction accuracy and robustness.

3. Proposed methodology

3.1. Theoretical foundation

To more effectively handle complex feature relationships in multivariate time series, we introduce an innovative group-wise processing mechanism. This mechanism not only adapts to differences between feature groups but also enhances the model's expressiveness while maintaining computational efficiency.

The group-wise processing mechanism is inspired by group convolution in computer vision, where different channel groups can capture diverse feature representations. In the time series context, this mechanism enables specialized pattern learning for correlated features, while significantly reducing computational complexity through parameter sharing, and enhancing model capacity through parallel processing. Furthermore, group-wise processing effectively isolates noise interference and prevents interactions between unrelated features, thereby enhancing model robustness. By configuring independent attention mechanisms for each feature group, the model can more precisely capture temporal dependencies within groups while maintaining independence between different feature groups, which is particularly important for handling multivariate time series data with distinct group characteristics (see Fig. 1). Given an input sequence $X = \{x_t\}_{t=1}^T \in \mathbb{R}^{T \times D}$, we partition the D -dimensional features into G groups, where each group processes $D_g = D/G$ features. This grouping strategy enables:

$$X^g \in \mathbb{R}^{T \times D_g}, g \in \{1, \dots, G\} \quad (1)$$

The feature grouping adopts a simple sequential division strategy based on fixed group size, aiming to maintain approximately 7 variables per group. Specifically, the 862 variables in the Traffic dataset are divided into 120 groups, the 321 variables in the Electricity dataset are divided into 45 groups, and the 21 variables in the Weather dataset are divided into 3 groups, all following their original sequence. While this straightforward sequential division method does not employ sophisticated clustering techniques, it offers advantages in terms of implementation simplicity and low computational overhead. Additionally, by maintaining the grouping relationship of variables in adjacent positions, it partially preserves the original structural characteristics of the data.

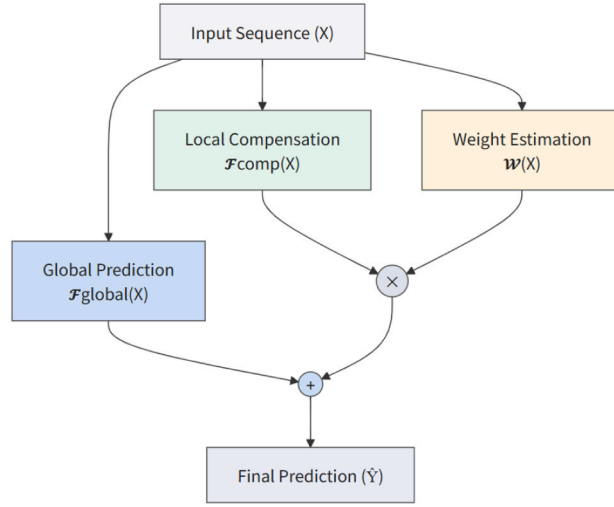


Fig. 1. Dual path learning mechanism.

3.2. Architecture components

This section details three core components of the model: the group-wise global trajectory estimator, adaptive compensation network, and dynamic weight modulation mechanism.

As shown in Fig. 1, these components form a dual-path learning framework comprising local and global branches. This design achieves more comprehensive temporal feature extraction and accurate prediction through parallel processing of features at different scales. In the framework, the input sequence (X) is separately fed into global prediction ($F_{\text{global}}(X)$) and local compensation ($F_{\text{comp}}(X)$) branches, while the weight estimation module ($W(X)$) dynamically adjusts the contribution of both branches, ultimately generating the final prediction ($\hat{Y}$) through weighted combination.

3.2.1. Group-wise global trajectory estimator

The global trajectory estimator employs an expansion-reduction mechanism to enhance feature representation:

$$H_t^g = \sigma(\mathcal{T}_g(X_{t-k:t}^g)W_g + b_g) \quad (2)$$

where $\mathcal{T}_g$ includes a dimension expansion operation:

$$\mathcal{T}_g(X) = \text{GELU}(XW_1^g)W_2^g \quad (3)$$

with $W_1^g \in \mathbb{R}^{d \times (d \cdot e)}$ and $W_2^g \in \mathbb{R}^{(d \cdot e) \times d}$, where e is the expansion factor.

The expansion factor enhances the model's capacity to capture complex temporal dependencies by creating a richer feature space for pattern extraction, thereby enabling multi-scale temporal feature learning. This design not only improves the model's expressiveness but also allows it to extract and learn feature patterns more effectively across different temporal scales.

3.2.2. Adaptive compensation network

The compensation mechanism is designed with multiple key considerations: incorporating error-aware feature refinement to identify and correct prediction biases, implementing adaptive scaling through temperature parameters to adjust feature sensitivity, and introducing dynamic importance weighting to balance the contributions of different features. These design elements work in concert to create a flexible and effective compensation framework.

For each group g , the compensation value is computed as:

$$C^g(X) = \sigma\left(\frac{F_{\text{comp}}^g(X)}{\tau^g(X)}\right) \cdot \mathcal{W}^g(X) \quad (4)$$

where F_{comp}^g learns residual patterns:

$$F_{\text{comp}}^g(X) = \text{MLP}(X - H_t^g) \quad (5)$$

The temperature parameter $\tau^g(X)$ controls feature sensitivity and is dynamically generated through a learnable network:

$$\tau^g(X) = \text{softplus}(\text{MLP}_\tau(X)) \quad (6)$$

The importance weight $\mathcal{W}^g(X)$ is used to adaptively adjust the compensation intensity:

$$\mathcal{W}^g(X) = \text{sigmoid}(\text{MLP}_w(X)) \quad (7)$$

The compensation network effectively captures prediction error patterns through residual learning (F_{comp}^g), while dynamically adjusting feature response sensitivity using temperature parameters (τ^g), and adaptively controlling compensation intensity through importance weights ($\mathcal{W}^g$). The entire compensation mechanism forms an end-to-end learnable framework that automatically adjusts compensation strategies based on different input features, improving the model's prediction accuracy and robustness.

3.2.3. Dynamic weight modulation

The weight modulation mechanism introduces learnable temperature parameters, which stems from the varying contributions of different feature groups to the final prediction. Through dynamic weight modulation, the model can adaptively adjust the importance of each feature group, thereby better handling the relationships between features and prediction requirements across different scenarios.

$$\mathcal{W}^g(X) = \phi\left(\frac{g^g(X)}{\tau}\right) \cdot \alpha^g(X) \quad (8)$$

where τ represents a learnable temperature parameter, ϕ denotes a soft-plus activation ensuring positive scaling, and $\alpha^g(X)$ is a group-specific importance weight.

This dynamic weight modulation mechanism has significant implications: it not only provides a flexible method for adjusting feature importance but also achieves smooth weight allocation through temperature parameters. The mechanism can automatically adjust the influence of feature groups based on input data characteristics, enhancing the model's adaptability and generalization performance while maintaining computational stability and interpretability.

3.3. Normalization strategy

To improve the model's training stability and generalization capability, we propose a comprehensive normalization and denormalization mechanism. The normalization process consists of statistical standardization followed by an optional affine transformation:

$$Z_{norm} = \text{Normalize}(X) \quad (9)$$

where the normalization process can be decomposed into several steps:

$$\mu = \frac{1}{N} \sum_{i=1}^N x_i \quad (10)$$

$$\sigma = \sqrt{\frac{1}{N} \sum_{i=1}^N (x_i - \mu)^2 + \epsilon} \quad (11)$$

$$\hat{x} = \gamma \cdot \frac{x - \mu}{\sigma} + \beta \quad (12)$$

where γ and β are learnable parameters for the affine transformation, and ϵ is a small constant for numerical stability.

The denormalization process reverses the normalization by applying the inverse operations:

$$x_{normalized} = \frac{\hat{x} - \beta}{\gamma + \epsilon^2} \quad (13)$$

$$X_{original} = x_{normalized} \cdot \sigma + \mu \quad (14)$$

The complete transformation pipeline can be expressed as:

$$Z_{out} = \mathcal{G}(\mathcal{H}(\text{Normalize}(F(X)), \theta)) \quad (15)$$

where F represents the initial feature transformation, $\mathcal{H}$ is the feature interaction function with learnable parameters θ , and $\mathcal{G}$ denotes the final transformation.

This mechanism effectively addresses the internal covariate shift problem while maintaining the model's representation power. The statistical moments (μ and σ) are computed across the feature dimensions and detached from the computational graph to ensure stable gradient propagation. The affine transformation parameters (γ and β) provide the model with the flexibility to learn optimal feature scales and shifts, while the denormalization process enables the recovery of features in their original scale when necessary.

The implementation supports multi-dimensional tensor inputs through dynamic dimension reduction:

$$\mu_d = \mathbb{E}_{d \in D} [X_d] \quad (16)$$

$$\sigma_d = \sqrt{\mathbb{V}_{d \in D} [X_d] + \epsilon} \quad (17)$$

where D represents the dimensions to be reduced during the normalization process.

The normalization and denormalization mechanism provides a comprehensive framework for feature processing, focusing on the impact of feature scale management on model convergence and generalization. Through dynamic computation of statistical quantities, optional affine transformations, and dynamic reduction strategies for multi-dimensional tensors, this mechanism offers a mathematically complete and flexible solution for feature transformation.

3.4. Implementation details

In our implementation, we employed PyTorch's ModuleList structure to achieve modular model management, where each submodule independently processes computational tasks for different feature groups. Through CUDA multi-stream parallel technology, we implemented cross-group parallel computation, optimizing the computational latency from the original $O(n)$ to $O(n/k)$, where k represents the number of parallel groups.

To address the vanishing gradient problem during training, we introduced the Residual Connection structure and designed a Dynamic Temperature Scaling mechanism. This mechanism adaptively adjusts the temperature parameter τ of the softplus layer based on the distribution characteristics of input features, effectively improving the model's generalization capability across different data distributions. Notably, the Sigmoid activation function constrains the range of local compensation, implementing the spatial decoupling concept mentioned in the paper. As shown in Fig. 2, the detailed network architecture illustrates the key components, including input normalization, multiple parallel processing branches with different activation functions (Sigmoid, Softplus, and GELU), and the integration of temperature scaling and residual connections.

4. Experimental results

4.1. Datasets introduction

In our study, we utilized nine diverse benchmark datasets to evaluate the performance of our proposed models. As shown in Table 1. These datasets cover a wide range of domains and exhibit varying characteristics in terms of the number of variables, time steps, and granularity. Table 1 provides an overview of these datasets, including the Electricity Transformer Temperature (ETT) datasets (ETTh1, ETTh2, ETTm1, and ETTm2), as well as datasets related to traffic flow, electricity consumption, exchange rates, weather conditions, and Influenza-Like Illness (ILI). The datasets differ significantly in their complexity, with the number of variables ranging from 7 to 862 and the number of time steps varying from 966 to 69,680. Additionally, the granularity of these datasets spans from 5 min intervals to weekly recordings, allowing us to test our models' capabilities across different temporal resolutions. Consistent with the benchmark datasets used in [9,39], our selection enables direct comparison with state-of-the-art time series forecasting approaches.

4.2. Experimental setup

4.2.1. Hardware environment and platform

All experiments were conducted on a Linux server equipped with an NVIDIA GeForce RTX 4060 GPU and 32 GB of memory. The server runs Ubuntu 24.04.1 LTS, using PyTorch 2.5.0 deep learning framework with CUDA 12.1. To ensure reproducibility, our code has been open-sourced on GitHub.

4.2.2. Training parameters

We adopted the following hyperparameter settings during model training:

- Batch size: 256 for ETT datasets, 32 for larger-scale datasets (such as Traffic and Electricity)
- Learning rate: Initially set to 0.01, halved after each epoch
- Optimizer: Adam optimizer with $\beta_1 = 0.9$, $\beta_2 = 0.999$, and weight decay coefficient of 0.01
- Training epochs: 10 epochs for all datasets
- Loss function: Mean Squared Error (MSE) as the primary loss function

4.2.3. Training strategy

To enhance model generalization and training stability, we implemented early stopping mechanism where training stops when validation loss shows no improvement for 3 consecutive epochs.

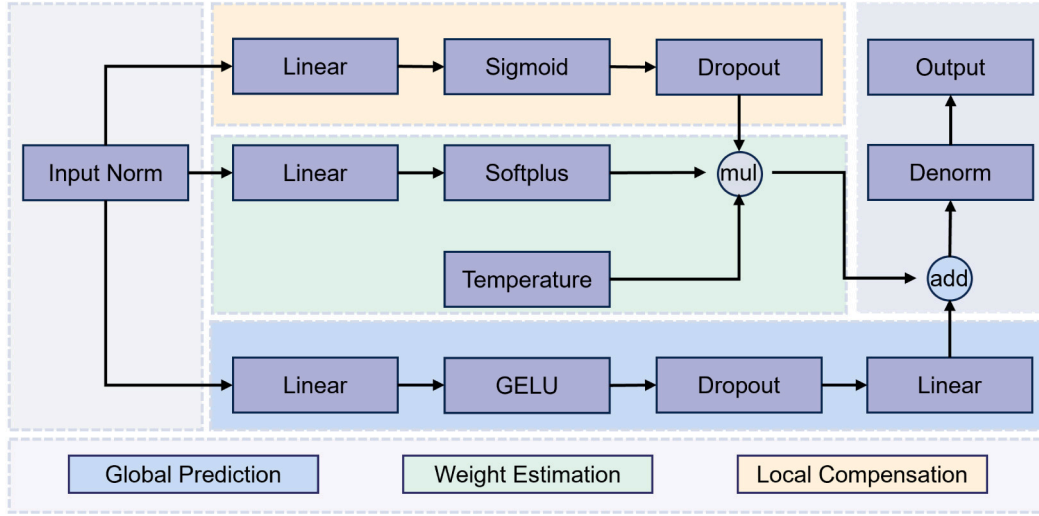


Fig. 2. Implementation architecture and key components of DualNet.

Table 1
The statistics of the nine benchmark datasets.

Datasets	Variates	Timesteps	Granularity
ETTh1&ETTh2	7	17,420	1 h
ETTm1&ETTm2	7	69,680	5 min
Traffic	862	17,544	1 h
Electricity	321	26,304	1 h
Exchange Rate	8	7,588	1 day
Weather	21	52,696	10 min
ILI	7	966	1 week

4.2.4. Evaluation metrics

We evaluated model performance using the following metrics:

- Mean Squared Error (MSE)
- Mean Absolute Error (MAE)

To ensure fair comparison, all baseline methods were tested under identical hardware conditions and data splits.

4.2.5. Baseline methods

- MICN [30] combines local features and global correlations, utilizing a multi-scale branch structure to model different potential temporal patterns.
- TimesNet [9] transforms 1D time series into 2D tensors based on multi-periodic observations, using 2D convolutions to extract temporal variation features.
- MSGNet [47] captures multi-scale inter-series correlations via frequency domain analysis and adaptive graph convolution.
- Crossformer [39] employs DSW embedding and a two-stage attention layer to capture dependencies across both time and variable dimensions.
- FEDformer [42] integrates seasonal-trend decomposition with frequency-enhanced Transformer modules to efficiently model long-term trends.
- TSMixer [21] adopts a lightweight MLP-Mixer architecture for forecasting by mixing features along both time and variable dimensions.
- DLinear [8] uses a simple one-layer linear model to demonstrate effectiveness for long-term time series forecasting.
- FITS [43] manipulates time series in the complex frequency domain through interpolation, achieving efficient forecasting with only about 10K parameters.

- TimeMixer [44] analyzes temporal variations with a multi-scale mixing perspective, leveraging PDM and FMM modules to aggregate microscopic seasonal and macroscopic trend information respectively.

4.3. Forecasting results

To evaluate the performance of our proposed models, we conducted extensive experiments on various time series datasets. These datasets have been extensively utilized for benchmarking and publicly available. Tables 2 and 3 present the comparison of long-term forecasting errors, measured by Mean Squared Error (MSE) and Mean Absolute Error (MAE), where lower values indicate better performance.

Table 2 presents a comparison of long-term forecasting errors for ETT (Electricity Transformer Temperature) datasets. ETT includes ETTh1, ETTh2, ETTm1, and ETTm2, all of which use a 96-hour historical observation window and predict future values for 96, 192, 336, and 720 h. The results show significant variations in performance across different methods, prediction horizons, and datasets.

Overall, DualNet demonstrates superior performance in most cases, particularly on ETTh1 and ETTm1 datasets, showing lower Mean Square Error (MSE) and Mean Absolute Error (MAE). This indicates DualNet's higher accuracy and stability in handling long-term prediction problems, adapting well to different datasets and prediction horizons.

Specifically, on the ETTh1 dataset, DualNet achieves an MSE of 0.362 for the 96-hour prediction horizon, reducing the error by approximately 9% compared to MICN's 0.398. Even in the most challenging 720-hour prediction, DualNet maintains good performance (MSE: 0.458), while Crossformer's error increases significantly to 0.750. On the ETTm2 dataset, DualNet achieves the lowest error (MSE: 0.172) for 96-hour predictions, showing a 8% reduction compared to TimesNet (MSE: 0.187), demonstrating clear performance advantages.

Other methods such as TSMixer, DLinear, and Crossformer, while performing well in certain scenarios, fail to maintain consistently high

Table 2
Comparison of long-term forecasting errors for ETT datasets.

Methods	Metric	ETTm1					ETTm2					ETTh1					ETTh2				
		96	192	336	720	Avg	96	192	336	720	Avg	96	192	336	720	Avg	96	192	336	720	Avg
MICN	MSE	0.316	0.363	0.408	0.481	0.392	0.179	0.307	0.325	0.502	0.328	0.398	0.430	0.460	0.491	0.445	0.332	0.422	0.447	0.442	0.411
	MAE	0.362	0.390	0.426	0.476	0.414	0.275	0.376	0.388	0.490	0.382	0.427	0.453	0.460	0.509	0.462	0.377	0.441	0.740	0.467	0.440
TimesNet	MSE	0.338	0.374	0.410	0.478	0.400	0.187	0.249	0.321	0.408	0.291	0.384	0.436	0.491	0.521	0.458	0.340	0.402	0.452	0.462	0.414
	MAE	0.375	0.387	0.411	0.450	0.406	0.267	0.309	0.351	0.403	0.333	0.402	0.429	0.469	0.500	0.450	0.374	0.414	0.452	0.468	0.427
MSGNet	MSE	0.319	0.376	0.417	0.481	0.398	0.177	0.247	0.312	0.414	0.288	0.390	0.442	0.480	0.494	0.452	0.328	0.402	0.435	0.417	0.396
	MAE	0.366	0.397	0.422	0.458	0.411	0.262	0.307	0.346	0.403	0.330	0.411	0.442	0.468	0.488	0.452	0.371	0.414	0.443	0.441	0.417
Crossformer	MSE	0.428	0.445	0.533	0.728	0.534	0.197	0.326	0.372	0.410	0.326	0.429	0.494	0.706	0.750	0.595	0.632	0.876	0.924	1.390	0.956
	MAE	0.444	0.468	0.519	0.655	0.522	0.321	0.375	0.421	0.448	0.391	0.440	0.482	0.625	0.689	0.559	0.547	0.663	0.702	0.863	0.694
FEDformer	MSE	0.379	0.426	0.445	0.543	0.448	0.203	0.269	0.325	0.421	0.305	0.376	0.420	0.459	0.506	0.440	0.358	0.429	0.496	0.463	0.437
	MAE	0.419	0.441	0.459	0.490	0.452	0.281	0.328	0.366	0.415	0.349	0.419	0.448	0.465	0.507	0.460	0.397	0.439	0.487	0.474	0.449
TSMixer	MSE	0.319	0.369	0.402	0.464	0.389	0.175	0.240	0.301	0.404	0.280	0.381	0.433	0.472	0.485	0.443	0.289	0.375	0.425	0.435	0.381
	MAE	0.358	0.384	0.406	0.442	0.398	0.258	0.230	0.338	0.398	0.306	0.391	0.420	0.441	0.471	0.431	0.338	<u>0.381</u>	0.435	0.449	0.403
DLinear	MSE	0.345	0.380	0.413	0.474	0.403	0.193	0.284	0.369	0.554	0.350	0.386	0.437	0.481	0.519	0.456	0.333	0.477	0.594	0.831	0.559
	MAE	0.372	0.389	0.413	0.453	0.407	0.292	0.362	0.427	0.522	0.401	0.400	0.432	0.459	0.516	0.452	0.387	0.476	0.541	0.657	0.515
FITS	MSE	0.351	0.392	0.424	0.485	0.413	0.181	0.246	0.306	0.407	0.285	0.381	0.434	0.474	0.464	0.438	0.290	0.375	0.414	0.419	0.375
	MAE	0.370	0.393	0.413	0.448	0.106	0.264	0.304	0.341	0.397	0.327	0.391	0.422	0.446	0.463	0.431	0.339	0.388	0.425	0.437	0.397
TimeMixer	MSE	0.320	0.361	<u>0.390</u>	<u>0.454</u>	0.381	0.175	0.237	0.298	0.391	0.275	0.375	0.429	0.484	0.498	0.447	0.289	0.372	<u>0.386</u>	<u>0.412</u>	<u>0.364</u>
	MAE	0.357	0.381	0.404	0.441	0.395	0.258	0.299	0.340	0.396	0.323	0.400	0.421	0.458	0.482	0.440	0.341	0.392	<u>0.414</u>	<u>0.434</u>	<u>0.395</u>
DualNet(Ours)	MSE	<u>0.315</u>	<u>0.358</u>	0.392	0.457	<u>0.381</u>	<u>0.172</u>	<u>0.235</u>	<u>0.293</u>	<u>0.391</u>	<u>0.273</u>	<u>0.362</u>	<u>0.415</u>	<u>0.452</u>	<u>0.458</u>	<u>0.422</u>	<u>0.286</u>	<u>0.367</u>	0.421	0.450	0.381
	MAE	<u>0.353</u>	<u>0.376</u>	<u>0.400</u>	<u>0.439</u>	<u>0.392</u>	<u>0.254</u>	<u>0.296</u>	<u>0.335</u>	<u>0.391</u>	<u>0.319</u>	<u>0.389</u>	<u>0.420</u>	<u>0.437</u>	<u>0.455</u>	<u>0.425</u>	<u>0.336</u>	0.389	0.431	0.456	0.403

performance across all contexts. This suggests that different methods may be better suited to specific data patterns or prediction requirements, indicating that studying these differences to optimize applicable prediction models will be an important direction for future work.

Table 3 presents a comparison of long-term forecasting errors across five different datasets: Electricity consumption, Exchange Rate, Traffic flow, Weather, and Influenza-Like Illness (ILI). The ILI dataset employs a 36-hour historical observation window with prediction horizons of 24, 36, 48, and 60 h, while the other four datasets use a 96-hour historical observation window with broader prediction ranges of 96, 192, 336, and 720 h.

Across these diverse datasets, DualNet continues to demonstrate robust prediction capabilities. In the electricity consumption dataset, DualNet achieves an MSE of 0.154 for 96-hour predictions, showing approximately 8.3% error reduction compared to TimesNet's 0.168. For the Traffic prediction task, the model particularly excels in long-term forecasting (720 h) with an MSE of 0.502, significantly outperforming other models such as Crossformer (MSE: 0.589) and FEDformer (MSE: 0.626).

Notably, DualNet shows exceptional performance on the ILI dataset, which operates on a distinctly different time scale. For 24-hour predictions, it achieves an MSE of 2.274, reducing the error by approximately 15% compared to MICN's 2.684. This consistent performance across different time scales further validates the model's adaptability and generalization capabilities. Even for the more challenging traffic flow and weather datasets, DualNet maintains relatively stable performance, highlighting its advantages in handling complex time series data.

Overall, through experiments on these five datasets with varying characteristics and prediction difficulties, DualNet demonstrates its potential as a universal time series prediction model. Compared to models optimized for specific domains, DualNet maintains strong predictive performance across various scenarios, which has significant implications for practical applications. These results also provide valuable references for designing more powerful time series prediction models in the future.

In both tables, we have underlined the best results to highlight the superior model performance across different datasets and forecasting horizons.

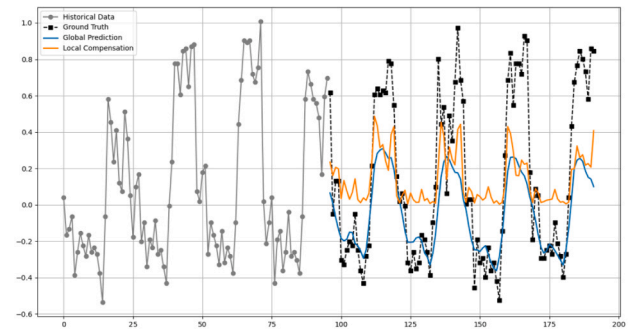


Fig. 3. Prediction decomposition.

4.4. Model validation analysis

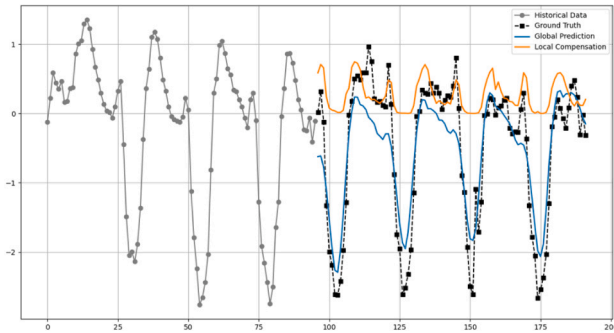
As shown in Fig. 3, the experimental results clearly demonstrate the prediction effectiveness of the DualNet architecture. The figure shows three key components: historical data (gray line), global prediction (black dashed line), and local compensation (orange solid line). From the time series performance, the global prediction path successfully captures the main trends and fluctuation patterns in the data, while the local compensation mechanism provides fine-tuned adjustments at key fluctuation points. Notably, during periods of intense data volatility (such as the 125–175 time interval), the model exhibits excellent adaptability: the global prediction path accurately tracks major trend changes, while the local compensation mechanism enhances overall prediction accuracy through minor fluctuation adjustments. This validates the effectiveness of the dual-path learning strategy proposed in the paper, which separately handles main trajectory estimation and fine-calibration through decoupling, thereby improving the model's ability to grasp complex temporal patterns while maintaining prediction stability.

As shown in Fig. 4, the experimental results clearly demonstrate the unique prediction mechanism of the DualNet architecture. The global prediction path (blue dashed line) exhibits distinct downward exploration characteristics, with this “aggressive” prediction mode effectively capturing the data's downward trends. Particularly in the time series' valley regions (such as the 125–175 time interval), the global

Table 3

Comparison of long-term forecasting errors on multivariate time series benchmarks.

Methods	Metric	Electricity					Exchange-Rate					Traffic					Weather					ILI				
		96	192	336	720	Avg	96	192	336	720	Avg	96	192	336	720	Avg	96	192	336	720	Avg	24	36	48	60	Avg
MICN	MSE	0.164	0.177	0.193	0.212	0.187	0.102	0.185	0.354	0.872	0.378	0.519	0.537	0.534	0.577	0.542	0.177	0.225	0.278	0.354	0.259	2.684	2.667	2.558	2.747	2.664
	MAE	0.269	0.285	0.304	0.321	0.295	0.235	0.332	0.469	0.727	0.441	0.309	0.315	0.313	0.325	0.316	0.218	0.259	0.297	0.348	0.281	1.112	1.068	1.052	1.110	1.086
TimesNet	MSE	0.168	0.184	0.198	0.220	0.192	0.107	0.226	0.367	0.964	0.416	0.593	0.617	0.629	0.640	0.620	0.172	0.219	0.280	0.365	0.259	2.371	1.972	2.238	2.027	2.139
	MAE	0.272	0.289	0.300	0.320	0.295	0.234	0.344	0.448	0.746	0.443	0.321	0.336	0.336	0.350	0.336	0.220	0.261	0.306	0.359	0.287	0.934	0.920	0.940	0.928	0.931
MSGNet	MSE	0.165	0.184	0.195	0.231	0.194	0.102	0.195	0.359	0.940	0.399	0.607	0.621	0.622	0.632	0.621	0.163	0.212	0.272	0.350	0.249	2.527	2.615	2.358	2.488	2.497
	MAE	0.274	0.292	0.302	0.332	0.300	0.230	0.317	0.436	0.738	0.430	0.392	0.399	0.396	0.396	0.395	0.212	0.254	0.299	0.348	0.278	1.020	1.007	0.972	1.016	1.004
Crossformer	MSE	0.219	0.231	0.246	0.280	0.244	0.256	0.470	1.268	1.767	0.940	0.522	0.530	0.558	0.589	0.550	0.158	0.206	0.272	0.398	0.259	4.849	4.709	4.942	4.974	4.869
	MAE	0.314	0.322	0.337	0.363	0.334	0.367	0.509	0.883	1.068	0.707	0.290	0.293	0.305	0.328	0.304	0.230	0.277	0.335	0.418	0.315	1.545	1.521	1.585	1.555	1.552
FEDformer	MSE	0.193	0.201	0.214	0.246	0.214	0.148	0.271	0.460	1.195	0.519	0.587	0.604	0.621	0.626	0.610	0.217	0.276	0.339	0.403	0.309	3.228	2.679	2.622	2.857	2.847
	MAE	0.308	0.315	0.329	0.355	0.327	0.278	0.380	0.500	0.841	0.500	0.366	0.373	0.383	0.382	0.376	0.296	0.336	0.380	0.428	0.360	1.260	1.080	1.078	1.157	1.144
TSMixer	MSE	0.204	0.218	0.235	0.271	0.232	0.245	0.452	1.223	1.705	0.906	0.508	0.515	0.542	0.573	0.535	0.152	0.198	0.263	0.385	0.250	4.726	4.589	4.815	4.850	4.745
	MAE	0.305	0.314	0.328	0.354	0.325	0.356	0.495	0.862	1.042	0.689	0.283	0.286	0.297	0.320	0.297	0.224	0.270	0.327	0.408	0.307	1.508	1.485	1.547	1.518	1.515
DLinear	MSE	0.197	0.196	0.209	0.245	0.212	0.088	0.176	0.313	0.839	0.354	0.650	0.598	0.605	0.645	0.625	0.196	0.237	0.283	0.345	0.265	2.398	2.646	2.614	2.804	2.616
	MAE	0.282	0.285	0.301	0.333	0.300	0.218	0.315	0.427	0.695	0.414	0.396	0.370	0.373	0.394	0.383	0.255	0.296	0.335	0.381	0.317	1.040	1.088	1.086	1.146	1.090
FITS	MSE	0.293	0.268	0.355	0.416	0.333	0.089	0.182	0.327	0.852	0.363	0.898	0.763	0.894	1.019	0.894	0.189	0.229	0.274	0.334	0.257	3.340	4.016	4.541	4.406	4.076
	MAE	0.401	0.378	0.452	0.498	0.432	0.210	0.303	0.414	0.696	0.406	0.572	0.522	0.608	0.646	0.587	0.248	0.288	0.326	0.371	0.308	1.299	1.453	1.554	1.517	1.456
TimeMixer	MSE	0.156	0.171	0.191	0.230	0.187	0.100	0.212	0.379	0.904	0.399	0.468	0.481	0.498	0.506	0.488	0.183	0.222	0.266	0.324	0.249	2.324	2.289	2.165	2.085	2.216
	MAE	0.249	0.265	0.283	0.315	0.278	0.222	0.326	0.442	0.715	0.426	0.294	0.305	0.324	0.312	0.309	0.241	0.280	0.317	0.361	0.300	1.080	0.931	0.908	0.909	0.957
DualNet(Ours)	MSE	0.154	0.168	0.185	0.225	0.183	0.085	0.176	0.327	0.886	0.369	0.465	0.473	0.487	0.502	0.482	0.161	0.210	0.264	0.344	0.245	2.274	2.103	2.121	2.023	2.130
	MAE	0.248	0.261	0.277	0.310	0.274	0.203	0.299	0.415	0.707	0.406	0.302	0.304	0.310	0.315	0.308	0.208	0.250	0.291	0.340	0.272	0.931	0.891	0.900	0.887	0.902

**Fig. 4.** Spatially decoupled dual-channel prediction diagram: Global downward exploration with residual compensation mechanism.

prediction path boldly tracks the data's downward movement. Meanwhile, the local compensation mechanism (orange solid line) focuses on repairing the upward residuals, bringing the prediction results back to the level of actual data (black dots) through precise compensatory adjustments. This clear division of upper and lower spaces allows both channels to perform their specialized roles: the global channel can freely explore downward trends without concerning about overall prediction accuracy, while the compensation channel focuses on repairing upward residuals. This decoupled design not only ensures stable training processes for both channels but also enhances the overall prediction accuracy.

4.5. Ablation experiment

To validate the effectiveness of our proposed components and better understand their contributions to the overall performance, we conduct comprehensive ablation studies by progressively removing key components from DualNet. We examine three critical aspects:

- The normalization and decoupling mechanism that forms the foundation of our dual-path architecture.
- The compensation mechanism that enables fine-grained calibration through residual learning.
- The temperature-scaled adaptive weighting scheme that dynamically balances prediction stability and refinement.

Table 4 presents the ablation study results of DualNet on ETT datasets, analyzing the impact of key components by progressively removing them from the model. The experiments compare various methods including standard normalization (standard norm), our proposed normalization strategy, variants without compensation mechanism (no comp), and temperature-scaled adaptive weighting (no temp) to demonstrate the effectiveness of each component.

The results demonstrate that the complete DualNet model achieves optimal performance across all datasets. Specifically, on the ETTh1 dataset, the complete model achieves an MSE of 0.315 for 96-hour predictions, significantly outperforming the version without normalization and decoupling (MSE: 0.323), confirming the importance of spatial decoupling strategy. On the ETTm1 dataset, the complete model's average MSE of 0.381 shows approximately 3.1% error reduction compared to the version without compensation mechanism (MSE: 0.393), indicating the effectiveness of the dual-path learning strategy's fine-grained calibration.

Notably, the removal of the temperature-scaled adaptive weighting mechanism also impacts model performance. For instance, the performance degradation in long-term predictions on the ETTh2 dataset without temperature scaling demonstrates that this dynamic compensation intensity adjustment mechanism helps the model achieve a better balance between prediction stability and adaptive optimization. This design proves particularly important when handling complex time series prediction tasks.

These experimental results thoroughly validate the necessity of DualNet's innovative components: the spatial decoupling strategy, compensation mechanism, and temperature-scaled adaptive weighting all contribute significantly to the model's overall performance. Furthermore, the results confirm the rationality of the proposed dual-channel architecture design, where components work synergistically to enhance the model's capability to capture complex temporal patterns.

4.6. Discussion of limitations

As illustrated in Fig. 5, our long-term prediction experiments revealed distinct periodic oscillation patterns in the model outputs. These oscillatory characteristics not only reflect the intrinsic dynamics of the system but also provide valuable insights into the model's long-term predictive behavior. This phenomenon can be attributed to several factors: First, the model might have overfitted to the periodic patterns

Table 4
Ablation study results of DualNet on ETT datasets.

Methods	Metric	ETTh1					ETTh2					ETTm1					ETTm2				
		96	192	336	720	Avg	96	192	336	720	Avg	96	192	336	720	Avg	96	192	336	720	Avg
standard norm	MSE	0.323	0.377	0.401	0.467	0.393	0.176	0.242	0.299	0.403	0.281	0.373	0.430	0.468	0.493	0.442	0.295	0.381	0.436	0.502	0.404
	MAE	0.372	0.414	0.416	0.454	0.414	0.285	0.321	0.422	0.472	0.357	0.402	0.433	0.488	0.521	0.461	0.382	0.431	0.452	0.611	0.469
no comp	MSE	0.317	0.370	0.393	0.458	0.385	0.173	0.237	0.293	0.395	0.275	0.366	0.422	0.459	0.483	0.433	0.289	0.374	0.427	0.492	0.396
	MAE	0.358	0.386	0.403	0.440	0.397	0.256	0.298	0.335	0.396	0.321	0.393	0.426	0.448	0.476	0.436	0.339	0.395	0.435	0.474	0.411
no temp	MSE	0.316	0.359	0.390	0.456	0.382	0.174	0.238	0.295	0.392	0.275	0.367	0.416	0.469	0.480	0.433	0.297	0.369	0.425	0.455	0.387
	MAE	0.357	0.378	0.401	0.439	0.394	0.257	0.298	0.334	0.391	0.320	0.391	0.419	0.445	0.473	0.432	0.346	0.392	0.434	0.458	0.408
DualNet	MSE	<u>0.315</u>	<u>0.358</u>	<u>0.392</u>	<u>0.457</u>	<u>0.381</u>	<u>0.172</u>	<u>0.235</u>	<u>0.293</u>	<u>0.391</u>	<u>0.273</u>	<u>0.362</u>	<u>0.415</u>	<u>0.452</u>	<u>0.458</u>	<u>0.422</u>	<u>0.286</u>	<u>0.367</u>	<u>0.421</u>	<u>0.450</u>	<u>0.381</u>
	MAE	<u>0.353</u>	<u>0.376</u>	<u>0.400</u>	<u>0.439</u>	<u>0.392</u>	<u>0.254</u>	<u>0.296</u>	<u>0.335</u>	<u>0.391</u>	<u>0.319</u>	<u>0.389</u>	<u>0.420</u>	<u>0.437</u>	<u>0.455</u>	<u>0.425</u>	<u>0.336</u>	<u>0.389</u>	<u>0.431</u>	<u>0.456</u>	<u>0.403</u>

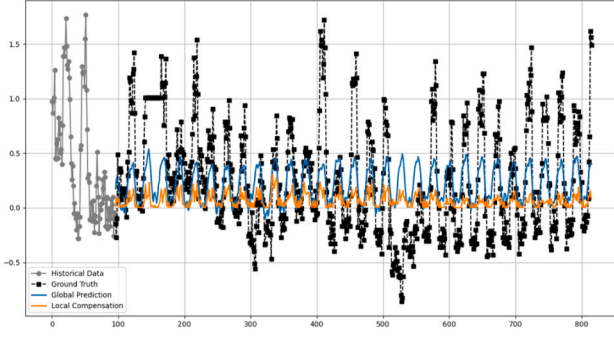


Fig. 5. Long-term prediction visualization analysis.

in the training data, leading to over-generalization of these patterns in long-term predictions. Second, as the prediction horizon extends to longer time scales, the model tends to degrade into a periodic prediction mode based on statistical features of historical data, rather than truly capturing the dynamic changes in the data. Notably, after time step 500, the predictions exhibit regular oscillations with relatively stable amplitude, indicating inherent limitations in the model's ability to handle long-term dependencies. While this periodic prediction pattern reflects the overall trend of the time series to some extent, it also reveals the model's difficulty in maintaining accurate modeling capabilities for complex nonlinear dynamics in long-term prediction tasks.

The periodic outputs observed in our long-term prediction experiments essentially reflect the issue of data scarcity. The current dataset lacks complete repetitive patterns, making it challenging for the model to learn genuine long-term evolution patterns. This phenomenon inspires us to reconsider that the core challenge in long-term prediction might not lie in the limitations of model architecture, but rather in the insufficiency of training samples. To validate this hypothesis, we conducted comparative experiments using noisy multi-frequency superimposed sinusoidal waveforms. As shown in Fig. 6, the selection of these composite waveforms is strategically significant: while maintaining fundamental periodic characteristics, they introduce real-world complexity through multi-frequency superposition and noise, thereby providing rich pattern information even with relatively limited samples. Through this approach, we can eliminate the impact of data scarcity and more accurately evaluate the model's intrinsic capability in long-term prediction tasks. This validation strategy will help us gain deeper insights into how sample sufficiency affects long-term prediction performance, providing important methodological guidance for future research.

As shown in Fig. 7, the model demonstrates exceptional long-term prediction performance on the synthetic multi-frequency superimposed sinusoidal waveform dataset. The predicted sequence (red curve) shows remarkable alignment with the true sequence (green curve), accurately capturing both the periodic variations and the local fluctuations introduced by superimposed noise. These results strongly support our core

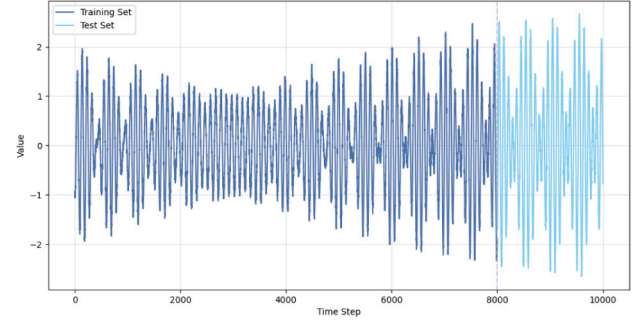


Fig. 6. Noisy multi-frequency superimposed sinusoidal waveforms.

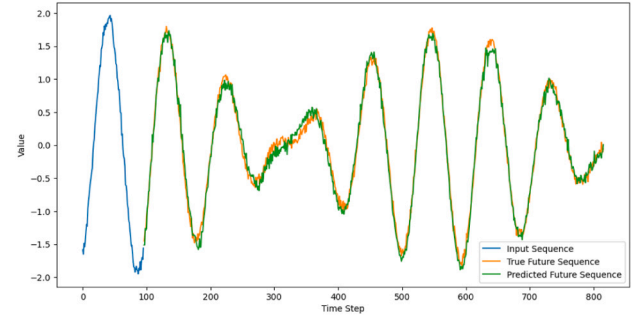


Fig. 7. Validation of long-term prediction capability using synthetic time series.

hypothesis: when training data contains sufficient pattern information, models can learn and predict long-term evolution patterns even with relatively limited samples. Notably, the predicted sequence maintains stable accuracy over an extended time horizon without significant error accumulation or prediction drift, further confirming the model architecture's inherent capability to handle long-term prediction tasks. This finding provides new insights for addressing long-term prediction challenges in real-world scenarios: the focus should be placed on obtaining and constructing training datasets with sufficient pattern information rather than excessive emphasis on model architecture improvements.

4.7. Time series similarity analysis

In previous experiments, we observed that long-term time series prediction performance is generally lower than short-term prediction, a phenomenon evident across various models. To gain deeper insight into the underlying reasons for this phenomenon, we conducted a time series similarity analysis. We hypothesized that if the subsequences within longer sequences exhibit lower similarity, this might make long-term prediction tasks inherently more challenging, as models need to capture more diverse and complex temporal patterns. We employed Matrix Profile analysis techniques to quantify time series similarity and verify this hypothesis.

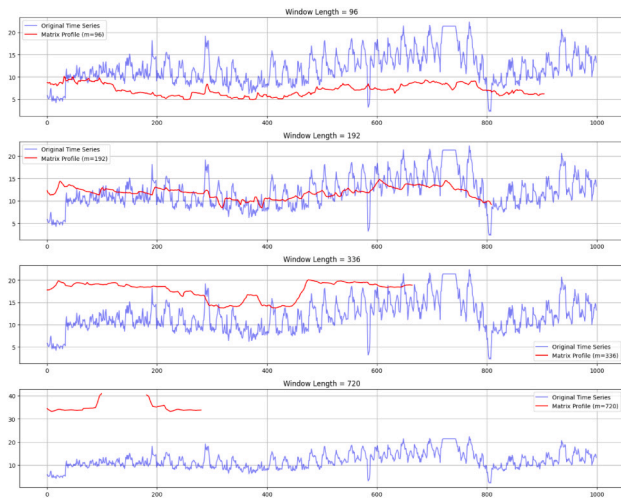


Fig. 8. Matrix profile analysis of ETTh1 dataset with different window lengths.

Matrix Profile [48] is an advanced data structure for time series analysis that characterizes anomalies and trends in time series by calculating distances between subsequences, commonly used for time series motif discovery. SCRIMP [49] utilizes randomized diagonal computation order and recursive updates of sliding dot products to efficiently compute and approximate the Matrix Profile. This study analyzed the Matrix Profile of the ETTh1 dataset across four sequence lengths: 96, 192, 336, and 720. As shown in the results, the mean value of the Matrix Profile gradually increases with window length, indicating decreased similarity between subsequences.

As shown in Fig. 8, the Matrix Profile analysis results reveal that the ETTh1 dataset exhibits significant characteristic differences across various time scales, which are manifested in the pattern structure and periodic variations of the time series. As the window length increases from 96 to 720, the mean value of the Matrix Profile progressively rises, intuitively reflecting a significant decrease in time series self-similarity at longer time scales. By observing the Matrix Profile curves across the four window lengths, we can see that shorter windows (96 and 192) display more local fluctuations and lower baseline values, while longer windows (336 and 720) not only show elevated overall levels but also exhibit smoother characteristics, suggesting that long sequence patterns have notably lower repeatability compared to short sequences.

The core challenge in long-term prediction lies in the insufficiency of samples, where it is difficult to find enough similar patterns even within the training data for neural networks to learn from. The Matrix Profile analysis clearly reveals this issue: as the prediction length increases, the number of similar patterns in the time series significantly decreases, directly limiting the neural network's ability to learn long-term dependencies from historical data. In other words, the poor performance of neural networks in long-term prediction tasks is largely not due to deficiencies in model design, but rather stems from the lack of sufficient long-term pattern samples in the training data, making it difficult for networks to accurately capture long-term variation patterns through traditional supervised learning approaches.

5. Conclusions

The DualNet architecture proposed in this study provides an effective solution for multivariate time series prediction through innovative decoupling design and dual-path learning strategy. Our research validates the effectiveness of decoupling principles in time series prediction and confirms that the dual-path learning strategy can effectively balance global pattern extraction and local detail compensation. Experimental results demonstrate that this method significantly enhances the

model's ability to capture complex temporal patterns while maintaining prediction stability. Although there are certain limitations in terms of computational complexity and parameter tuning, these findings provide important theoretical references and technical support for the field of time series prediction. Future research can focus on model efficiency optimization, automatic parameter tuning, and cross-domain applications to further enhance the practical value of this method. The potential demonstrated by decoupling design and adaptive mechanisms in handling complex prediction tasks also provides valuable insights for research in other related fields.

CRedit authorship contribution statement

Huashun Li: Writing – original draft, Software, Methodology. **Weimin Wu:** Supervision, Conceptualization. **Wei Chen:** Writing – original draft, Validation. **Mei Zhang:** Supervision, Resources, Funding acquisition.

Funding information

This project was supported by the National Natural Science Foundation of China (Grant No. 51874010 and No. 52374154)

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Mei Zhang reports financial support was provided by National Natural Science Foundation of China. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

References

- [1] José F. Torres, Dalil Hadjout, Abderrazak Sebaa, Francisco Martínez-Álvarez, Alicia Troncoso, Deep learning for time series forecasting: a survey, *Big Data* 9 (1) (2021) 3–21.
- [2] Bryan Lim, Stefan Zohren, Time-series forecasting with deep learning: a survey, *Phil. Trans. R. Soc. A* 379 (2194) (2021) 20200209.
- [3] Ming Jin, Huan Yee Koh, Qingsong Wen, Daniele Zamboni, Cesare Alippi, Geoffrey I. Webb, Irwin King, Shirui Pan, A survey on graph neural networks for time series: Forecasting, classification, imputation, and anomaly detection, *IEEE Trans. Pattern Anal. Mach. Intell.* (2024).
- [4] Xu Liu, Yutong Xia, Yuxuan Liang, Junfeng Hu, Yiwei Wang, Lei Bai, Chao Huang, Zhenguang Liu, Bryan Hooi, Roger Zimmermann, Largest: A benchmark dataset for large-scale traffic forecasting, in: *Advances in Neural Information Processing Systems*, vol. 36, 2023, pp. 75354–75371.
- [5] Junyi Tang, Simin An, Yuanwei Liu, Yong Su, Jin Chen, M2ast: Mlp-mixer-based adaptive spatial-temporal graph learning for human motion prediction, *Multimedia Syst.* 30 (4) (2024) 206.
- [6] Fabio Giampaolo, Federico Gatta, Edoardo Prezioso, Salvatore Cuomo, Mengchu Zhou, Giancarlo Fortino, Francesco Piccialli, ENCODE-ensemble neural combination for optimal dimensionality encoding in time-series forecasting, *Inf. Fusion* 100 (2023) 101918.
- [7] Fan Yang, Ke Yan, Ning Jin, Yang Du, Multiple households energy consumption forecasting using consistent modeling with privacy preservation, *Adv. Eng. Inform.* 55 (2023) 101846.
- [8] Ailing Zeng, Muxi Chen, Lei Zhang, Qiang Xu, Are transformers effective for time series forecasting? *proc. AAAI conf. artif. intell.* 37 (9) (2023) 11121–11128.
- [9] Haixu Wu, Tengge Hu, Yong Liu, Hang Zhou, Jianmin Wang, Mingsheng Long, Timesnet: Temporal 2d-variation modeling for general time series analysis, 2022, arXiv preprint [arXiv:2210.02186](https://arxiv.org/abs/2210.02186).

- [10] T. Song, Z. Wang, P. Xie, N. Han, J. Jiang, D. Xu, A novel dual path gated recurrent unit model for sea surface salinity prediction, *J. Atmos. Ocean. Technol.* 37 (2) (2020) 317–325.
- [11] Z. Lu, B. Li, C. Fu, J. Wu, L. Xu, S. Jia, H. Zhang, Remaining useful life prediction method based on dual-path interaction network with multiscale feature fusion and dynamic weight adaptation, *Actuators* 13 (10) (2024) 413.
- [12] H. Chi, Y. Lu, C. Xie, W. Ke, B. Chen, Spatio-temporal attention based collaborative local-global learning for traffic flow prediction, *Eng. Appl. Artif. Intell.* 139 (2025) 109575.
- [13] D. Ge, B. Liu, Y. Cheng, Z. Zhu, Rethinking decomposition for time series forecasting: Learning from temporal patterns, *Expert Syst. Appl.* (2025) 128044.
- [14] Y. Chen, K. Li, C.K. Yeo, K. Li, Global-local feature learning via dynamic spatial-temporal graph neural network in meteorological prediction, *IEEE Trans. Knowl. Data Eng.* (2024).
- [15] Zhicheng Cui, Wenlin Chen, Yixin Chen, Multi-scale convolutional neural networks for time series classification, 2016, arXiv preprint [arXiv:1603.06995](https://arxiv.org/abs/1603.06995).
- [16] Jin Fan, Ke Zhang, Yipian Huang, Yifei Zhu, Baiping Chen, Parallel spatio-temporal attention-based TCN for multivariate time series prediction, *Neural Comput. Appl.* 35 (18) (2023) 13109–13118.
- [17] Guokun Lai, Wei-Cheng Chang, Yiming Yang, Hanxiao Liu, Modeling long-and short-term temporal patterns with deep neural networks, in: *The 41st International ACM SIGIR Conference on Research & Development in Information Retrieval*, 2018, pp. 95–104.
- [18] Yifan. Zhang, Rui. Wu, Sergiu.M. Dascalu, Frederick.C.Harris. Jr., A novel extreme adaptive gru for multivariate time series forecasting, *Sci. Rep.* 14 (1) (2024) 2991.
- [19] Haoyi Zhou, Shanghang Zhang, Jieqi Peng, Shuai Zhang, Jianxin Li, Hui Xiong, Wancai Zhang, Informer: Beyond efficient transformer for long sequence time-series forecasting, *Proc. AAAI Conf. Artif. Intell.* 35 (12) (2021) 11106–11115.
- [20] Haixu Wu, Jiehui Xu, Jianmin Wang, Mingsheng Long, Autoformer: Decomposition transformers with auto-correlation for long-term series forecasting, *Adv. Neural Inf. Process. Syst.* 34 (2021) 22419–22430.
- [21] E. Vijay, Arindam Jati, Nam Nguyen, Gift Sinthong, Jayant Kalagnanam, Tsmixer: Lightweight mlp-mixer model for multivariate time series forecasting, in: *ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 2023.
- [22] A.R. Troncoso-García, María Martínez-Ballesteros, Francisco Martínez-Álvarez, Alicia Troncoso, A new approach based on association rules to add explainability to time series forecasting models, *Inf. Fusion* 94 (2023) 169–180.
- [23] Qiang Guo, Lexin Fang, Ren Wang, Caiming Zhang, Multivariate time series forecasting using multiscale recurrent networks with scale attention and cross-scale guidance, *IEEE Trans. Neural Netw. Learn. Syst.* (2023).
- [24] Site Mo, Haoxin Wang, Bixiong Li, Songhai Fan, Yuankai Wu, Xianggen Liu, TimeSQL: Improving multivariate time series forecasting with multi-scale patching and smooth quadratic loss, *Inform. Sci.* 671 (2024) 120652.
- [25] Shun-Yao Shih, Fan-Keng Sun, Hung-yi Lee, Temporal pattern attention for multivariate time series forecasting, *Mach. Learn.* 108 (2019) 1421–1441.
- [26] Srihari Athiyarath, Mousumi Paul, Srivatsa Krishnaswamy, A comparative study and analysis of time series forecasting techniques, *SN Comput. Sci.* 1 (3) (2020) 175.
- [27] Pablo Montero-Manso, Rob J. Hyndman, Principles and algorithms for forecasting groups of time series: Locality and globality, *Int. J. Forecast.* 37 (4) (2021) 1632–1653.
- [28] Michael P. Clements, Do professional forecasters believe in the Phillips curve? *Int. J. Forecast.* 40 (3) (2024) 1238–1254.
- [29] Hamed Jabbari Asl, Eiji Uchibe, Online estimation of objective function for continuous-time deterministic systems, *Neural Netw.* 172 (2024) 106116.
- [30] Huiqiang Wang, Jian Peng, Feihu Huang, Jince Wang, Junhui Chen, Yifei Xiao, Micn: Multi-scale local and global context modeling for long-term series forecasting, in: *The Eleventh International Conference on Learning Representations*, 2023.
- [31] Minhao Liu, Ailing Zeng, Muxi Chen, Zhijian Xu, Qiuxia Lai, Lingna Ma, Qiang Xu, Scinet: Time series modeling and forecasting with sample convolution and interaction, *Adv. Neural Inf. Process. Syst.* 35 (2022) 5816–5828.
- [32] Emadeldeen Eldele, Mohamed Ragab, Zhenghua Chen, Min Wu, Xiaoli Li, Tslanet: Rethinking transformers for time series representation learning, 2024, arXiv preprint [arXiv:2404.08472](https://arxiv.org/abs/2404.08472).
- [33] Mingyue Cheng, Jiqian Yang, Tingyue Pan, Qi Liu, Zhi Li, ConvtimeNet: A deep hierarchical fully convolutional model for multivariate time series analysis, 2024, arXiv preprint [arXiv:2403.01493](https://arxiv.org/abs/2403.01493).
- [34] David Salinas, Valentin Flunkert, Jan Gasthaus, Tim Januschowski, DeepAR: Probabilistic forecasting with autoregressive recurrent networks, *Int. J. Forecast.* 36 (3) (2020) 1181–1191.
- [35] Farah Shahid, Aneela Zameer, Muhammad Muneeb, A novel genetic LSTM model for wind power forecast, *Energy* 223 (2021) 120069.
- [36] Shahzad Muzaffar, Afshin Afshari, Short-term load forecasts using LSTM networks, *Energy Procedia* 158 (2019) 2922–2927.
- [37] Hossein Abbasimehr, Mostafa Shabani, Mohsen Yousefi, An optimized model using LSTM network for demand forecasting, *Comput. Ind. Eng.* 143 (2020) 106435.
- [38] Gerald Woo, Chenghao Liu, Doyen Sahoo, Akshat Kumar, Steven Hoi, Etsformer: Exponential smoothing transformers for time-series forecasting, 2022, arXiv preprint [arXiv:2202.01381](https://arxiv.org/abs/2202.01381).
- [39] Yunhao Zhang, Junchi Yan, Crossformer: Transformer utilizing cross-dimension dependency for multivariate time series forecasting, in: *The Eleventh International Conference on Learning Representations*, 2023.
- [40] Yong Liu, Tengge Hu, Haoran Zhang, Haixu Wu, Shiyu Wang, Lintao Ma, Mingsheng Long, iTransformer: Inverted transformers are effective for time series forecasting, 2023, arXiv preprint [arXiv:2310.06625](https://arxiv.org/abs/2310.06625).
- [41] Jiaxin Gao, Wenbo Hu, Yuntian Chen, Client: Cross-variable linear integrated enhanced transformer for multivariate long-term time series forecasting, 2023, arXiv preprint [arXiv:2305.18838](https://arxiv.org/abs/2305.18838).
- [42] Tian Zhou, Ziqing Ma, Qingsong Wen, Xue Wang, Liang Sun, Rong Jin, Fedformer: Frequency enhanced decomposed transformer for long-term series forecasting, in: *International Conference on Machine Learning*, 2022, pp. 27268–27286.
- [43] Zhijian Xu, Ailing Zeng, Qiang Xu, FITS: Modeling time series with 10k parameters, 2023, arXiv preprint [arXiv:2307.03756](https://arxiv.org/abs/2307.03756).
- [44] Shiyu Wang, Haixu Wu, Xiaomeng Shi, Tengge Hu, Huakun Luo, Lintao Ma, James Y. Zhang, Jun Zhou, Timemixer: Decomposable multiscale mixing for time series forecasting, 2024, arXiv preprint [arXiv:2405.14616](https://arxiv.org/abs/2405.14616).
- [45] Lu Han, Xu-Yang Chen, Han-Jia Ye, De-Chuan Zhan, SOFTS: Efficient multivariate time series forecasting with series-core fusion, 2024, arXiv preprint [arXiv:2404.14197](https://arxiv.org/abs/2404.14197).
- [46] S. Nie, S. An, Q. Peng, Y. Su, COLLAR: Combating low-rank temporal latent representation for high-dimensional multivariate time series prediction using dynamic Koopman regularization, 2025, researchsquare preprint.
- [47] Wanlin Cai, Yuxuan Liang, Xianggen Liu, Jianshuai Feng, Yuankai Wu, Ms-gnet: Learning multi-scale inter-series correlations for multivariate time series forecasting, *Proc. AAAI Conf. Artif. Intell.* 38 (10) (2024) 11141–11149.
- [48] Chin-Chia Michael Yeh, Yan Zhu, Liudmila Ulanova, Nurjahan Begum, Yifei Ding, Hoang Anh Dau, Diego Furtado Silva, Abdullah Mueen, Eamonn Keogh, Matrix profile I: all pairs similarity joins for time series: a unifying view that includes motifs, discords and shapelets, in: *IEEE 16th International Conference on Data Mining, ICDM, 2016*, pp. 1317–1322.
- [49] Yan Zhu, Chin-Chia Michael Yeh, Zachary Zimmerman, Kaveh Kamgar, Eamonn Keogh, Matrix profile XI: SCRIMP++: time series motif discovery at interactive speeds, in: *IEEE International Conference on Data Mining, ICDM, 2018*, pp. 837–846.



Huashun Li, male, born in 1999, received his Bachelor's degree in Automation from the School of Electrical and Information Engineering at Anhui University of Science and Technology, Huainan, China. He is currently pursuing a Ph.D. degree in Mining Mechatronic Engineering at the same university. His research interests include time series prediction, clustering analysis, and interpretability of neural networks. He has participated in two National Natural Science Foundation of China projects and published more than 10 papers in important academic journals and conferences, including 3 SCI-indexed papers. He has received the National Scholarship for Graduate Students and other academic honors.



Weimin Wu, male, born in 1974, is currently serving as the Dean of the School of Electrical and Information Engineering at Anhui University of Science and Technology and a doctoral supervisor, while also serving as an Associate Editor of IEEE Transactions on Industrial Electronics. Recognized as an Outstanding Academic Leader of Shanghai, Shuguang Scholar, and Pujiang Scholar, his main research interests include renewable energy generation technology, power electronic devices and applications, energy storage, and microgrid control technology. He has led several major research projects, including International Cooperation Projects under the National Key R&D Program and National

Natural Science Foundation projects, and was awarded the Second Prize of Shanghai Natural Science Award (as the first accomplisher). Having published over 200 academic papers, obtained 18 invention patents, and authored one English monograph, he has been consecutively named an Elsevier Highly Cited Chinese Scholar from 2020 to 2023. Throughout his career, he has supervised more than 100 master's students and 6 doctoral students.



Wei Chen, male, born in 1993, Bachelor degree in Electronic Information Technology and Instrument, School of Computer Science and Engineering, Anhui University of Science and Technology, Hefei. He is currently pursuing a Ph.D. in intelligent electromechanical systems at Anhui University of Science and Technology. His research interests include digital twins, RUL prediction, multimodal data processing, etc. He participated in 3 National Natural Science Foundation projects and published more than 5 papers in important academic journals and conferences, including 2 papers included in sci. He has won many academic honors such as national Scholarship for postgraduate students.



Mei Zhang, female, born in November 1979, is currently serving as an Associate Professor and Master's supervisor at the School of Electrical and Information Engineering, Anhui University of Science and Technology, where she obtained her Master's degree in Engineering in 2005. She teaches several undergraduate courses including Circuit Theory, Analog Electronics, Digital Electronics, and Electronic Circuit Innovation Practice. Her research interests focus on intelligent detection and fault diagnosis, and IoT applications. She has led key provincial research projects and participated in multiple National Natural Science Foundation projects, with a total funding of over 680,000 yuan. Having published more than 40 academic papers, including 6 SCI-indexed papers, 4 EI-indexed papers, and numerous core Chinese journal publications, she has also obtained three invention patents as the first inventor. She has received multiple awards including the Provincial Teaching New Star Award of Anhui Province (2019) and has won several teaching excellence awards at university level. Throughout her career, she has demonstrated outstanding achievements in both teaching and research, particularly in the development of innovative teaching methods and student practical training.